

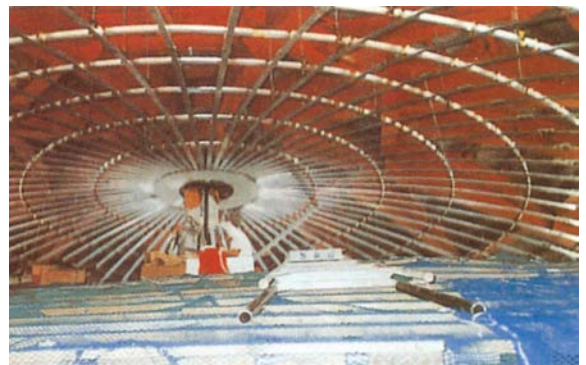
KIRII

CEILING ***SUSPENSION SYSTEM***

As a comprehensive manufacturer of building materials, we dedicated to developing our products for wider application.



Gypsum Ceiling Suspension System



Dome Ceiling Suspension System

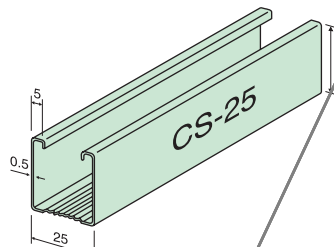
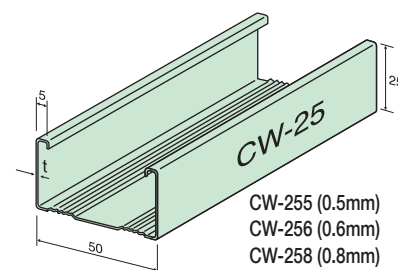
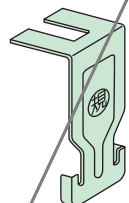
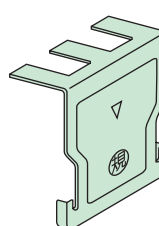
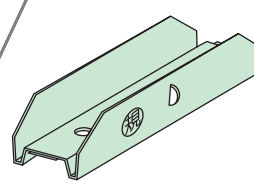
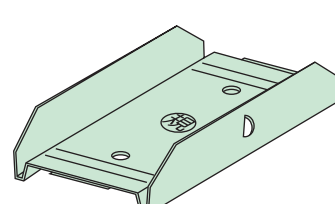
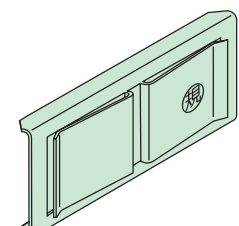
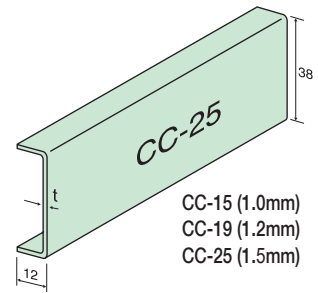
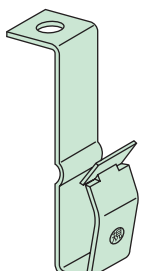
CONSTRUCTION MATERIALS FOR CEILINGS

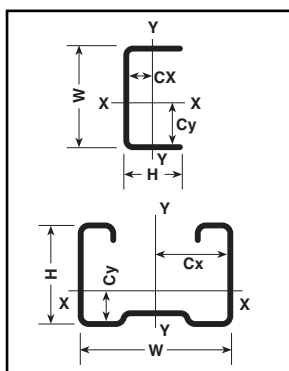
Making safe and pleasant living space means that the invisible portions must be strong. We have changed with the progress of the modern construction and our construction materials for wall and ceilings are now forming, the heart of the modern construction projects. "Providing only the finest to our customers" is the guiding tenet which follows our products from the quality-control at the factory through our computerized product distribution system.



Curve Ceiling Suspension System

HD-25 (HEAVY DUTY) SYSTEM **KIRII**

KIRII-I KIRII-II KIRII-III KIRII-IV KIRII-URAYASU			HD-25 ceiling members carry carved mark of product code and accessories have a mark (規)		
 CS-25 Single Furring			 CW-255 (0.5mm) CW-256 (0.6mm) CW-258 (0.8mm) Double Furring		
 CS-Clip (1000pcs/box)			 CW-Clip (500pcs/box)		
 CS-Joint (300pcs/box)			 CW-Joint (160pcs/box)		
 3/8" Suspension Bolt Shown as an option for vertical hanger only, other types of hangers may be adopted, all subject to specification/approval			 CC-Joint (300pcs/box)		
			 CC-15 (1.0mm) CC-19 (1.2mm) CC-25 (1.5mm) Carrying Channel		
			 CC-Hanger (300pcs/box)		
			 JIS Certification		



Lightweight ceiling members with their sectional properties

Product Code	W x H x T	A (cm ²)	I _x (cm ²)	I _y (cm ⁴)	Z _x (cm ³)	Z _y (cm ³)	(kg/m)
CC-15	38 x 12 x 1.0	0.600	1.210	0.073	0.637	0.078	0.473
CC-19	38 x 12 x 1.2	0.697	1.360	0.840	0.716	0.089	0.567
CC-25	38 x 12 x 1.5	0.870	1.702	0.102	0.895	0.112	0.687
CW-255	50 x 25 x 0.5	0.575	0.486	2.357	0.291	0.943	0.440
CW-256	50 x 25 x 0.6	0.670	0.561	2.766	0.336	1.106	0.530
CW-258	50 x 25 x 0.8	0.890	0.712	3.618	0.420	1.447	0.702

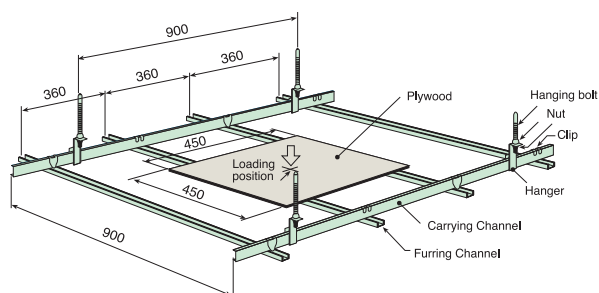
HD-25 (HEAVY DUTY) SYSTEM KIRII

The HD-25 (Heavy Duty) Ceiling Suspension System has the following features:

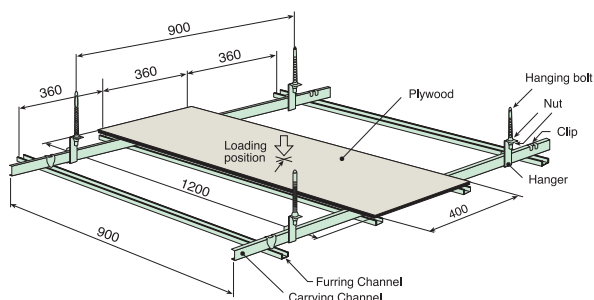
- JIS A 6517 and ASTM C645 standards.
- Furring channels are capable of withstanding the superimposed load of 50 kgf.
- Rust proof, as the whole suspension system is made of hot-dip galvanized steel with a protective zinc coating of Z18, 180g/m² (G60).
- Can create a monolithic and seamless ceiling, beautiful and close to living environment.
- Easy to install and adjust for an accurate ceiling level.
- Totally non-combustible, lightweight and durable.
- Can be accessible by use of access panel.
- Ideal for use in hotels, airports, apartments and industrial buildings.



Downward Loading Test

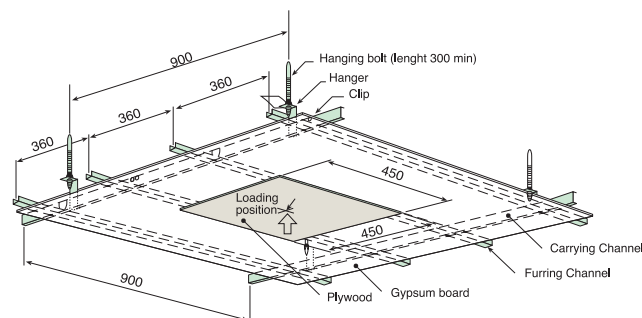


1. Downward Loading Test for Furring Channels (Type 25). As shown above, the two furring channels at the center should have deflection values measured after being loaded continuously for 5 minutes on to a piece of plywood board of 450 x 450 x 15mm placed on them. The furring channels were tested with the loading of 50kgf (490.33 N). The deflection values were derived from each channel and the larger value was taken. The residual deflection was measured after unloading.



2. Downward Loading Test for Carrying Channels (CC-25). As shown above, the two carrying channels at the center point between the suspension bolts should have the maximum deflection values measured after being loaded with 75 kgf (735.50 N) continuously for 5 minutes onto a piece of plywood board of 1200 x 400 x 30mm placed on them. The deflection values were derived from each carrying channel, and the larger value was taken. The residual deflection value was also measured after unloading.

Upward Loading Test

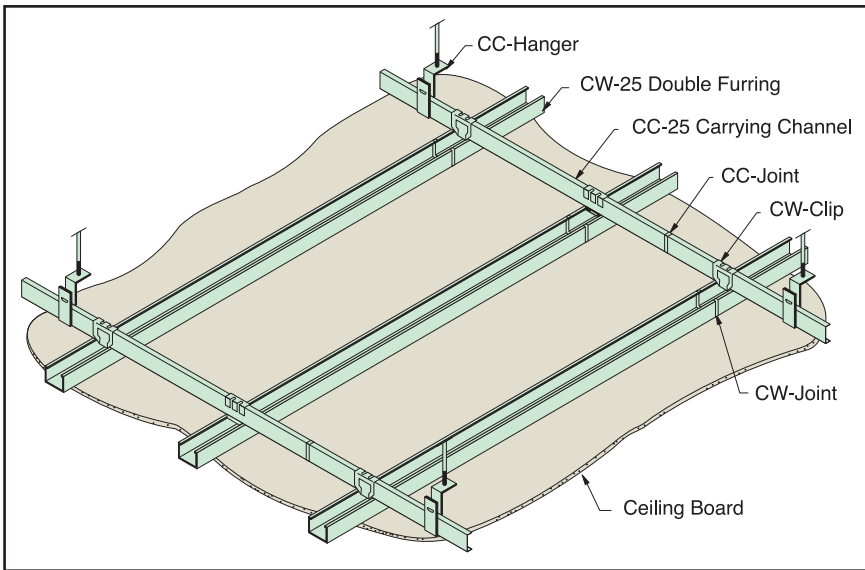


Ceiling materials were duly assembled as shown above, with additional installation of a piece of gypsum board 9mm thick prescribed as JIS A 6901 ceiling material. In the center of the ceiling, a piece of plywood of 450 x 450 x 15mm was placed underneath, and the upward successive loading tests with 30kgf (294.2 N) were applied for 5 minutes. The maximum deflection values were measured in the center of each furring channel, and the larger value was taken.

Item \ Type		Type 25	Test needed
Zinc volume galvanized (1)		Minimun 120g/m ² (both faces)	Zinc Volume
Stability of component in shape	Lateral bend	2L/1000mm or less	Component's Lateral bend & Camber
	Camber		
Load sustainable	Downward load	Furring Channel Maximum deflection 10mm or less Maximun residual deflection 2mm or less	Downward loading
		Carrying Channel Maximum deflection 5mm or less Maximum residual deflection 2mm or less	Downward loading
	Upward load		Maximundeflection 5mm or less

Note: (1) Zinc volume galvanized shall be in accordance with JIS G 3302.

HD-25 (HEAVY DUTY) SYSTEM KIRII

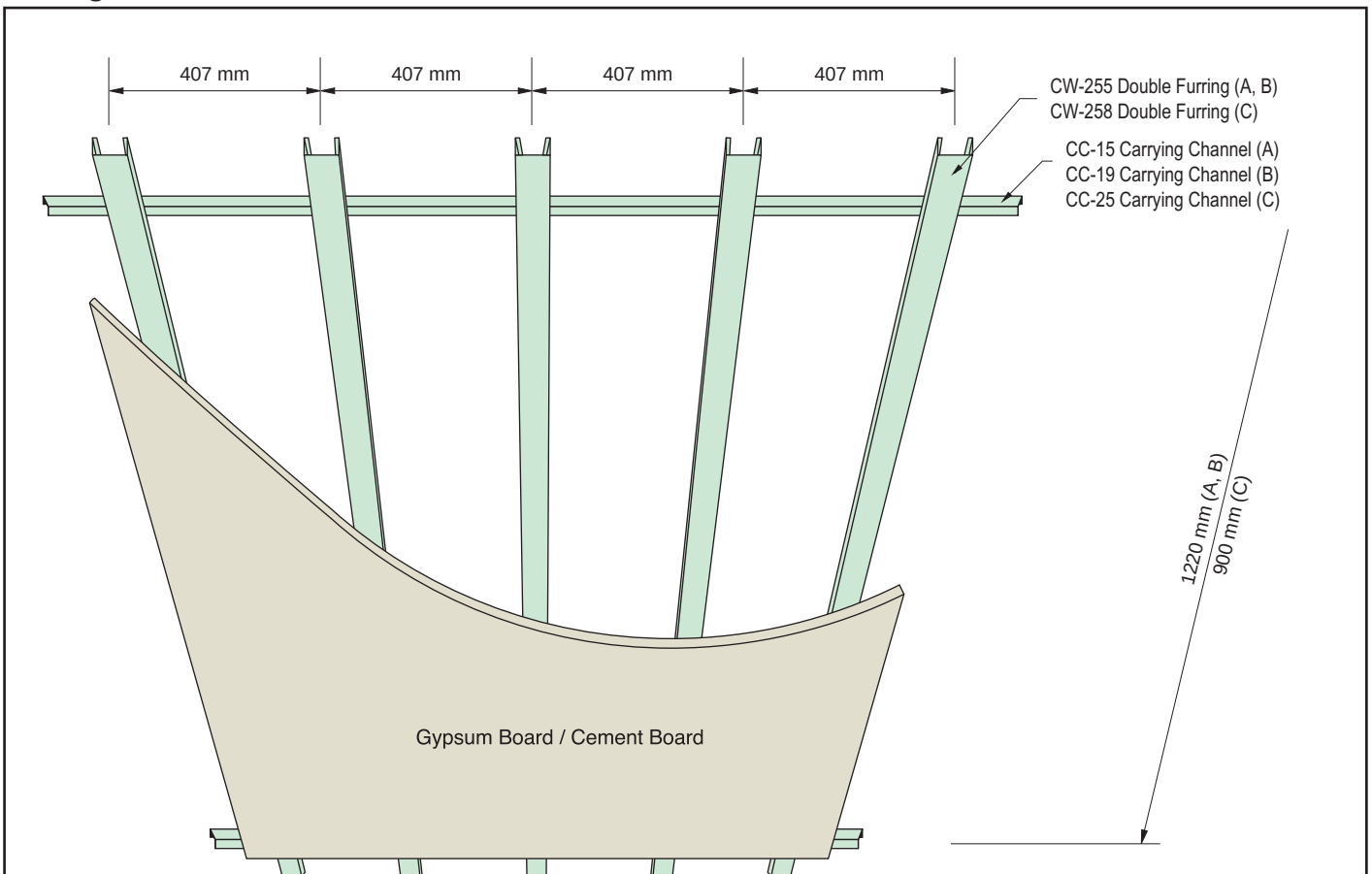


HD-25 (Heavy Duty) Ceiling Suspension System is suitable for fixing all kinds of ceiling boards such as gypsum board, fibre cement board and calcium silicate board etc.

According to the weight and size of the board, some of the members shall adjust the spacing to suit. The following ceiling patterns with uniform distributed load are recommended for installation.

For external application to resist wind load, please contact to our office for further information.

Ceiling Frame Patterns



Pattern	Uniform Load
A	20 kg/m ²
B	25 kg/m ²
C	50 kg/m ²

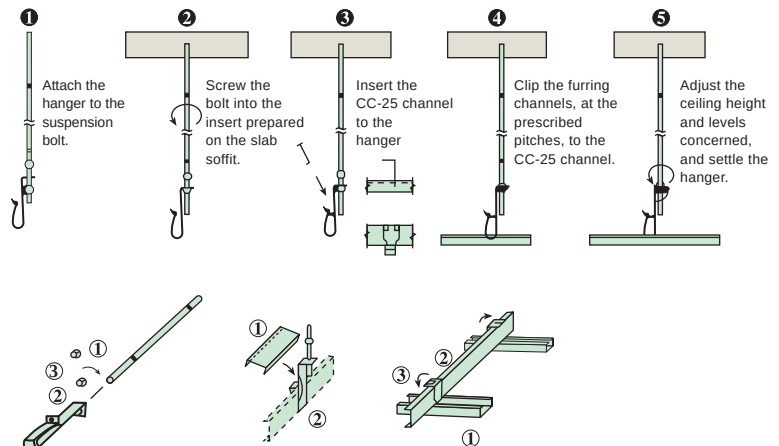
Notes:

1. allowable deflection limit of L/360
2. comply with ASTM C645 and JIS A6517

HD-25 (HEAVY DUTY) SYSTEM KIRII

WORKING PROCEDURE

- (1) Level Marking
Mark the ceiling height on the surrounding walls according to the reference line.
- (2) Installing suspension hangers
Fit the hanger lightly to the lower part of the suspension bolt.
- (3) Installing suspension bolts
Install the suspension bolt to the insert prepared on the slab soffit.
- (4) Installing CC-25 channels
Install the CC-25 channel to the hanger.
- (5) Determining the level
String and level the CC-25 channel according to the mark above, and tighten the hanger nut.
- (6) Installing furring channels
Clip the furring channels to the CC-25 channel according to the backing allotment. Clips must as a rule alternate with the sides of the CC-25 channel.
- (7) Installing ceiling board
Fix the board by bugle head screws to the furring channels at 300mm (12") max.



Architectural Specification of HD-25 Ceiling Suspension System for Gypsum Board & Fiber Cement Board Ceilings

A) SUSPENSION SYSTEM

- a) The suspension system for gypsum board shall be KIRII HD-25 (Heavy Duty) ceiling suspension system, frame pattern A with a uniform distributed load of 20kg/m².
- b) The framing members shall be made of hot-dip galvanized steel with a protective zinc coating of Z18, 180g/m² (G60). The cross sectional dimension of CC-25 carrying channel and CW-256 double furring shall be 38 x 12 x 1.5mm and 50 x 25 x 0.6mm respectively.
- c) As an alternative proposal, the supporting calculation shall be submitted and approved by the architect to ensure that the structural is adequate to sustain the ceiling load.

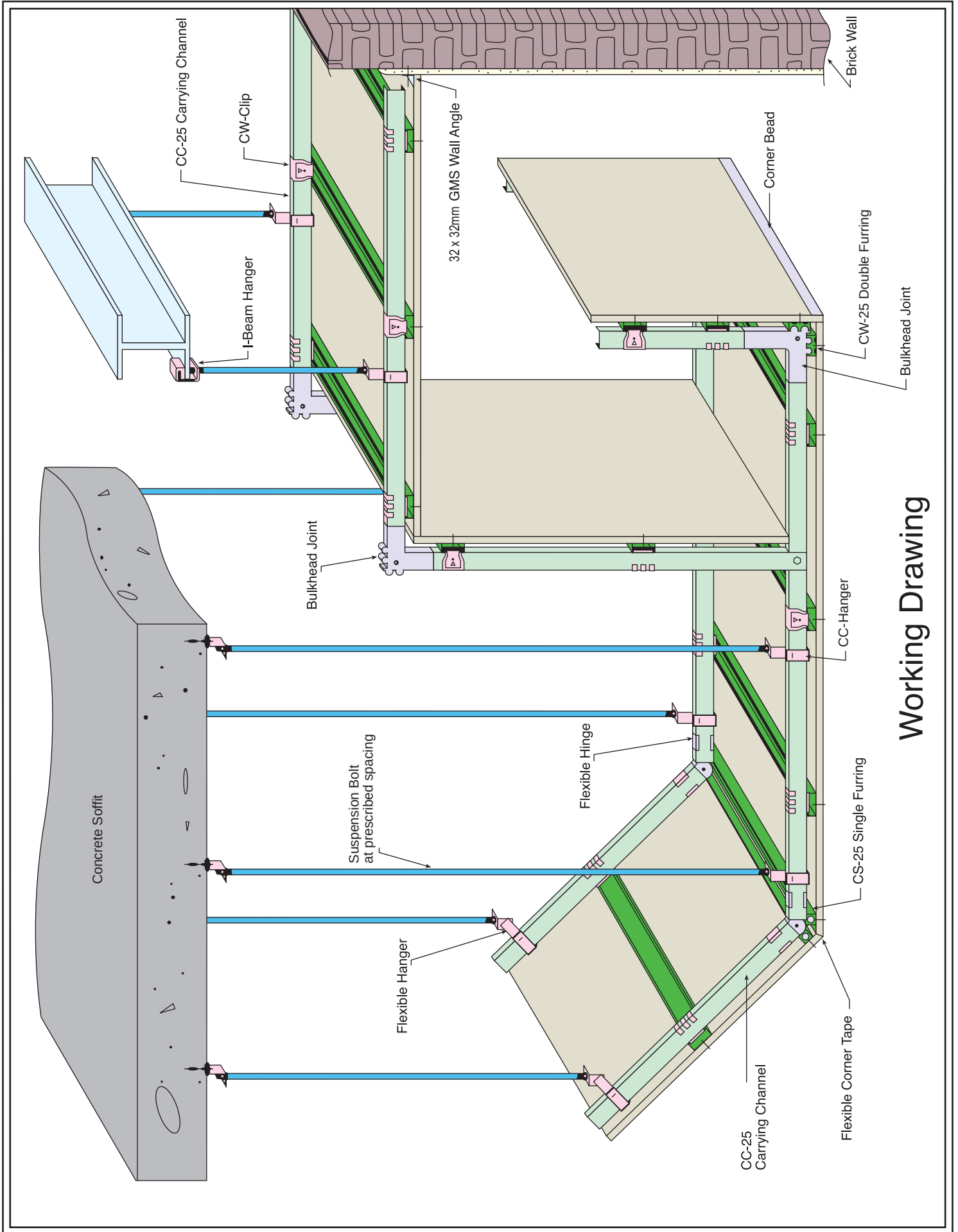
B) CEILING BOARDS

- Gypsum board shall be “Gyproc” M2TECH moisture and mold resistant, 1/2” (12.7mm) thick with tapered edges and complying with ASTM C1396 standard.
- Fiber cement board shall be “Villaboard”, 9mm thick with recessed edges for flush jointing and complying with ISO 8336-1993, AS 2908.2-1992 and ASTM C1185 standards.

C) INSTALLATION

- a) Installation of the ceiling suspension hangers shall be by suitably fixing to the underside of the slab soffit. Spacing of suspension hangers to be at 1220mm max. in either directions and at perimeter of ceiling as required.
- b) The carrying channel must be placed at 1220mm max. shall be hung to the lower part of the suspension hangers by means of galvanized steel fixings and nuts.
- c) The double furring at a spacing of 407mm max. shall be fixed to the carrying channel with double clips at each intersection. Double clips must as a rule alternate with the sides of the carrying channel.
- d) Rigid and aligned system for the fixing of ceiling board must be provided with separated pieces of carrying channel and double furring to be jointed by channel joint and double joint respectively.
- e) Extra hangers to be provided where opening are required for light fittings and access panels.
- f) All components must be well fixed and the suspension system must be correctly aligned and accurately leveled before attaching the ceiling board.
- g) The gypsum board shall be fixed by S25 screws at 300mm max. to the double furring with long edges of the gypsum board placed parallel to the double furring. (Fastener for Villaboard shall be HardiDrive screw, refer to James Hardie Ceiling Linings for fixing details).
- h) Finish of all joints and screw heads with joint tape and ready-mix jointing compound must be in accordance with manufacturer’s instruction which shall be approved by the architect and sand smooth.
- i) Plastering and painting works can be proceeded.

HD-25 (HEAVY DUTY) SYSTEM **KIRII**



Working Drawing



株式会社 桐井製作所

KIRII CONSTRUCTION MATERIAL CO., LTD.

HEAD OFFICE:

IMPERIAL TOWER 18TH FLOOR, UCHISAIWAI-CHO 1-1-1, CHIYODAKU,
TOKYO 100-0011, JAPAN

TEL : (81-3) 3539-6710

FAX : (81-3) 3539-6690

香港桐井有限公司

KIRII (HONG KONG) LIMITED

No. 9, DAI FU STREET, TAI PO INDUSTRIAL ESTATE, TAI PO, N.T., HONG KONG

TEL : (852) 2797 2026

FAX : (852) 2341 2618

E-mail address : info@kirii.com.hk